The former is called one-way classified data and the latter is called two-way classified data. Applications of ANOVA technique to these kinds of data are discussed in the following sections.

### 3.3.1 One-way ANOVA

ANOVA is a statistical technique used to determine whether differences exist among three or more population means.

In one-way ANOVA the effect of one factor on the mean is tested. It is based on independent random samples drawn from $k$ – different levels of a factor, also called treatments.

The following notations are used in one-way ANOVA. The data can be represented in the following tabular structure.

**Data representation for one-way ANOVA**

| Treatments | | | | | Total |
|---|---|---|---|---|---|
| Treatment 1 | $x_{11}$ | $x_{12}$ | … | $x_{1n_1}$ | $x_{1.}$ |
| Treatment 2 | $x_{21}$ | $x_{22}$ | … | $x_{2n_2}$ | $x_{2.}$ |
| ⋮ | ⋮ | ⋮ | … | ⋮ | ⋮ |
| Treatment $k$ | $x_{k1}$ | $x_{k2}$ | … | $x_{kn_k}$ | $x_{k.}$ |

$x_{ij}$ - the $j^{th}$ sample value from the ith treatment, $j = 1, 2, \ldots, n_i$, $i = 1, 2, \ldots, k$

$k$ - number of treatments compared.

$x_{i.}$ - the sample total of $i^{th}$ treatment.

$n_i$ - the number of observations in the $i^{th}$ treatment.

$$\sum_{i=1}^{k} n_i = n$$

The total variation in the observations $x_{ij}$ can be split into the following two components

i) variation between the levels or the variation due to different bases of classification, commonly known as treatments.

ii) The variation within the treatments *i.e.* inherent variation among the observations within levels.

Causes involved in the first type of variation are known as assignable causes. The causes leading to the second type of variation are known as chance or random causes.

The first type of variation that is due to assignable causes, can be detected and controlled by human endeavor and the second type of variation that is due to chance causes, is beyond the human control.

### 3.3.2 Test Procedure

Let the observations $x_{ij}$, $j = 1, 2, \ldots, n_i$ for treatment $i$, be assumed to come from $N(\mu_i, \sigma^2)$ population, $i = 1, 2, \ldots, k$ where $\sigma^2$ is unknown.



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